

WIWYNN CONFIDENTIAL

Sealion

Project Code: B1082301
Project Name: Sealion
PCB Number: 23X21
PCB Version: SA
PCB Name: Sensor control board
PCB Size: 175mm x 40mm
PCB Layer: 8L
PCB Thickness: 1.57mm+/- 10% mm

<Variant Name>

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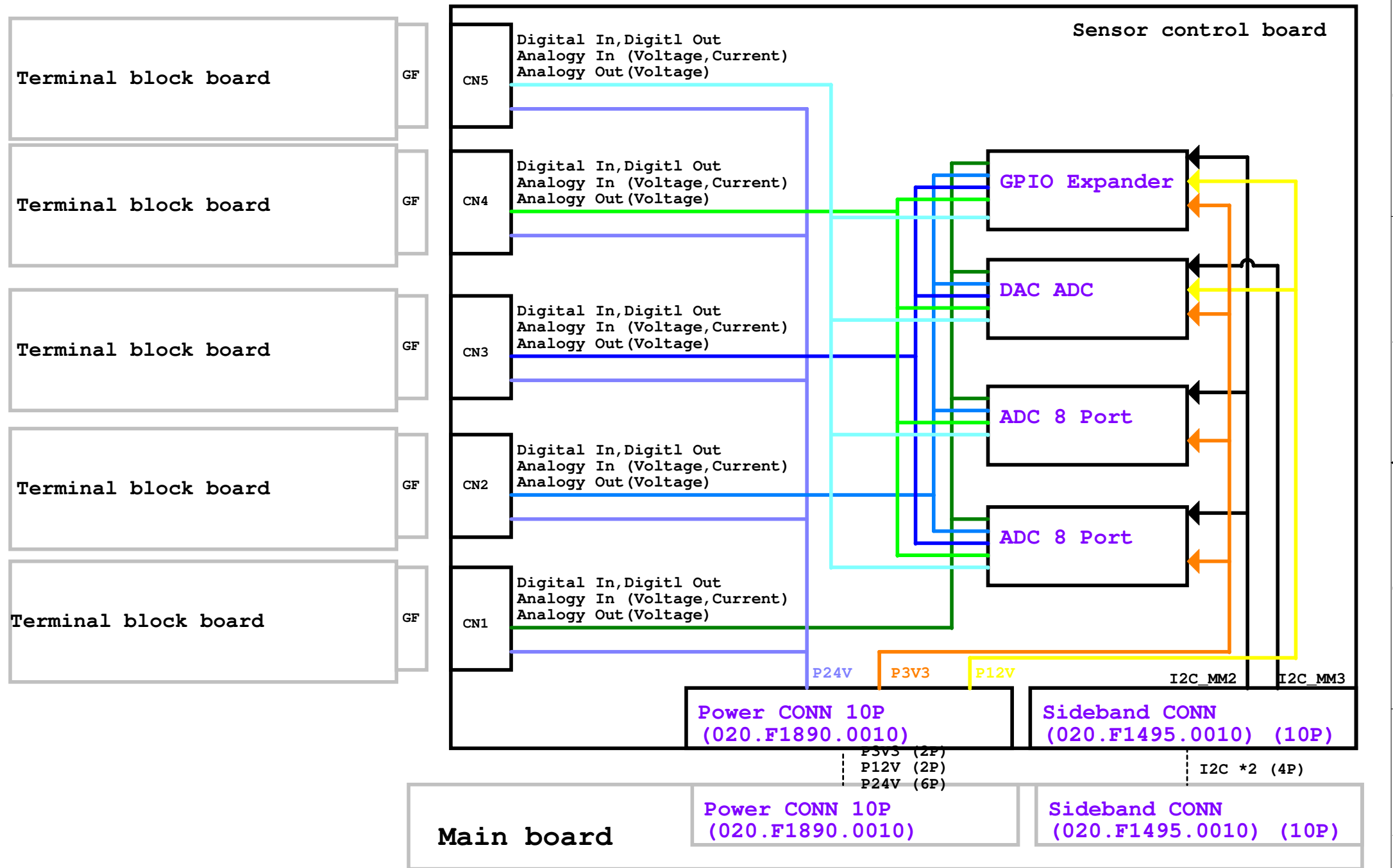
TABLE OF CONTENTS

NOPOP=no populated

562R2F-GP
562 = 562 ohm, (2K2R means 2.2K ohm)
2 = size 0402
F = 1% tolerance
GP= Green Part (RoHS)
Wiwynn RC size code as below:
1 = 0201
2= 0402
3= 0603
5= 0805
6= 1206
Wiwynn R tolerance code as below:
D=0.5%
F= 1%
J= 5%

SCD1U10V2KX-5GP
D1U = 0.1uF (2D2U means 2.2uF)
10Voltage (6D3V means 6.3V)
2 = size 0402, K tolerance
K=tolerance
[Wiwynn C code as below:]
G=2%
J=5%
K=10%
M=20%
X=temp characteristics
[(RoHS) Wiwynn C Series/Temp]
N=NPO
X=X7R/X5R
Y=Y5V
-5=different symbol/customer
GP= Green Part (RoHS)

SYSTEM BLOCK DIAGRAM



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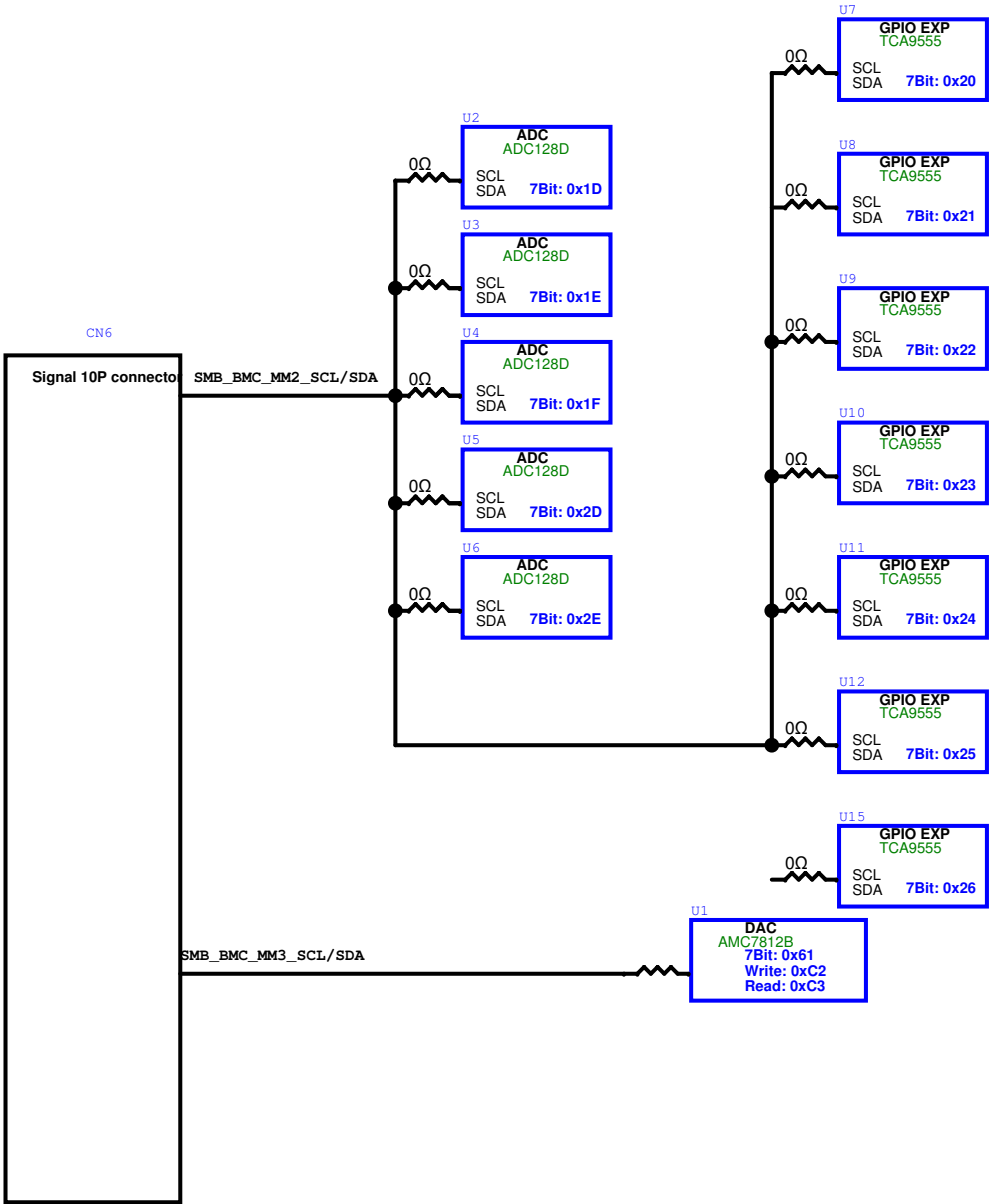
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I2C BLOCK DIAGRAM



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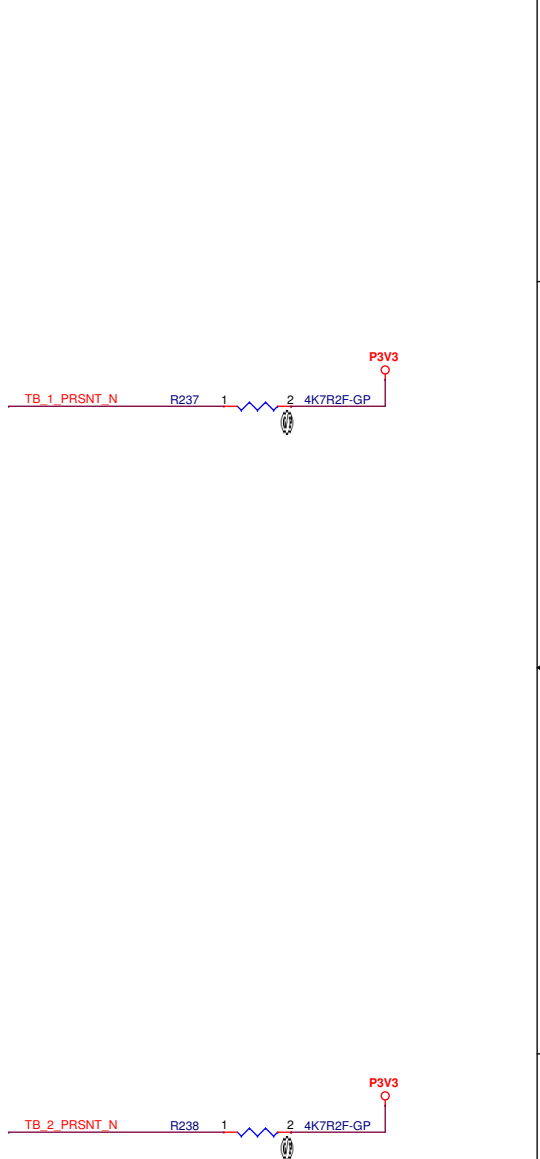
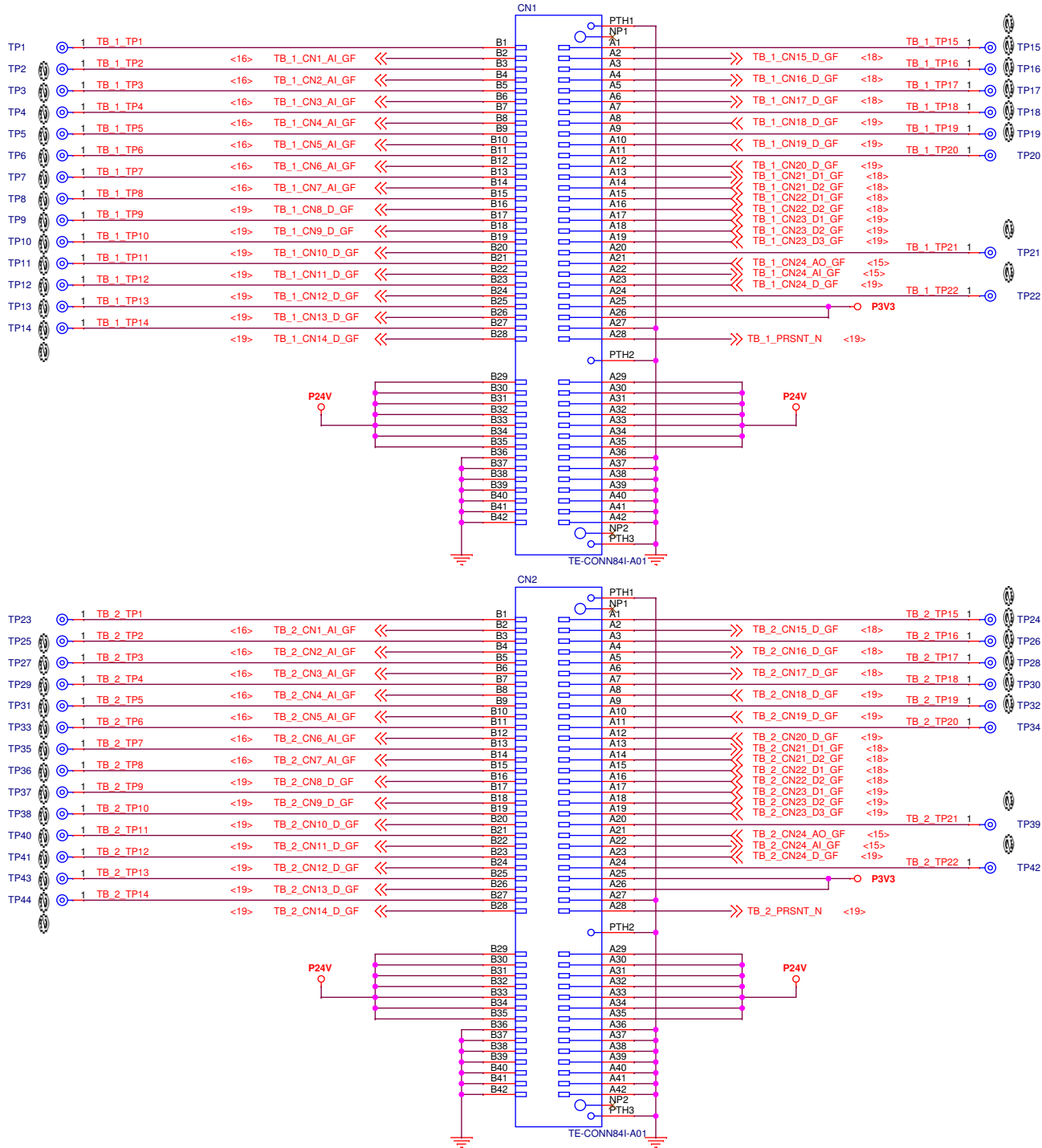
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Conn 1/2 to terminal block board

A2	EMER BTN
A4	SW LED1 N
A6	SW LED2 N
A8	PWR CTL1 N
A10	PWR CTL2 N
A12	LC PWR CTL1 N
A13	LC RX N
A14	LC PWR CTL2 N
A15	MOTOR CTL CW
A16	MOTOR CTL CCW
A17	ON OFF VAL CTL1 N
A18	ON OFF VAL CTL2 N
A19	ON OFF VAL PWR CTL N
A21	BALL VAL AO
A22	BALL VAL AI
A23	BALL VAL PWR CTL N
A27	GF TP23
B2	VOL SEN AI
B4	RES SEN01 AI
B6	RES SEN02 AI
B8	RES SEN03 AI
B10	CUR SEN01 AI
B12	CUR SEN02 AI
B14	CUR SEN03 AI
B16	SW1 N
B18	SW2 N
B20	SW3 N
B22	SW4 N
B24	SW5 N
B26	SW6 N
B28	SW7 N



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TB_CONN_CN1_CN2

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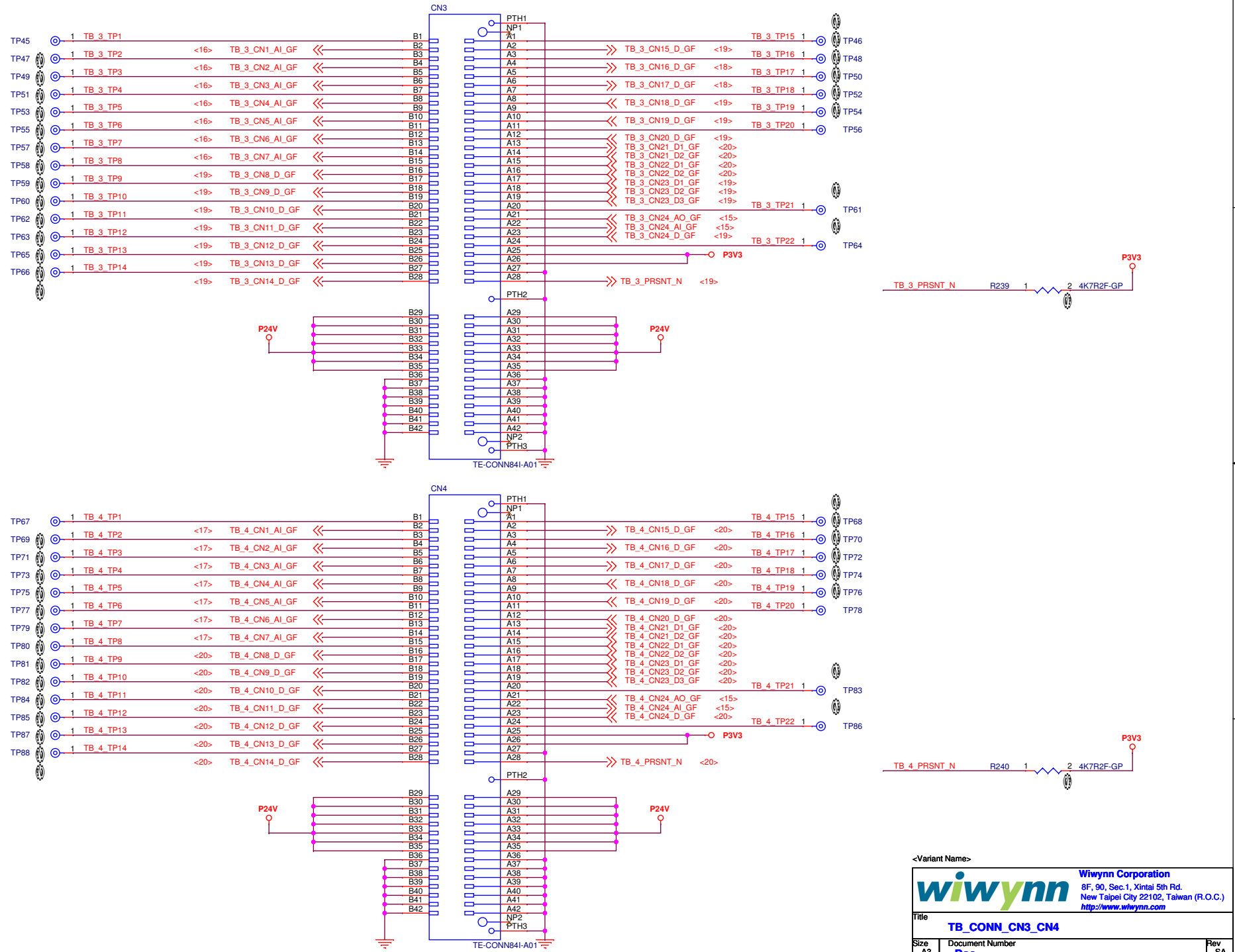
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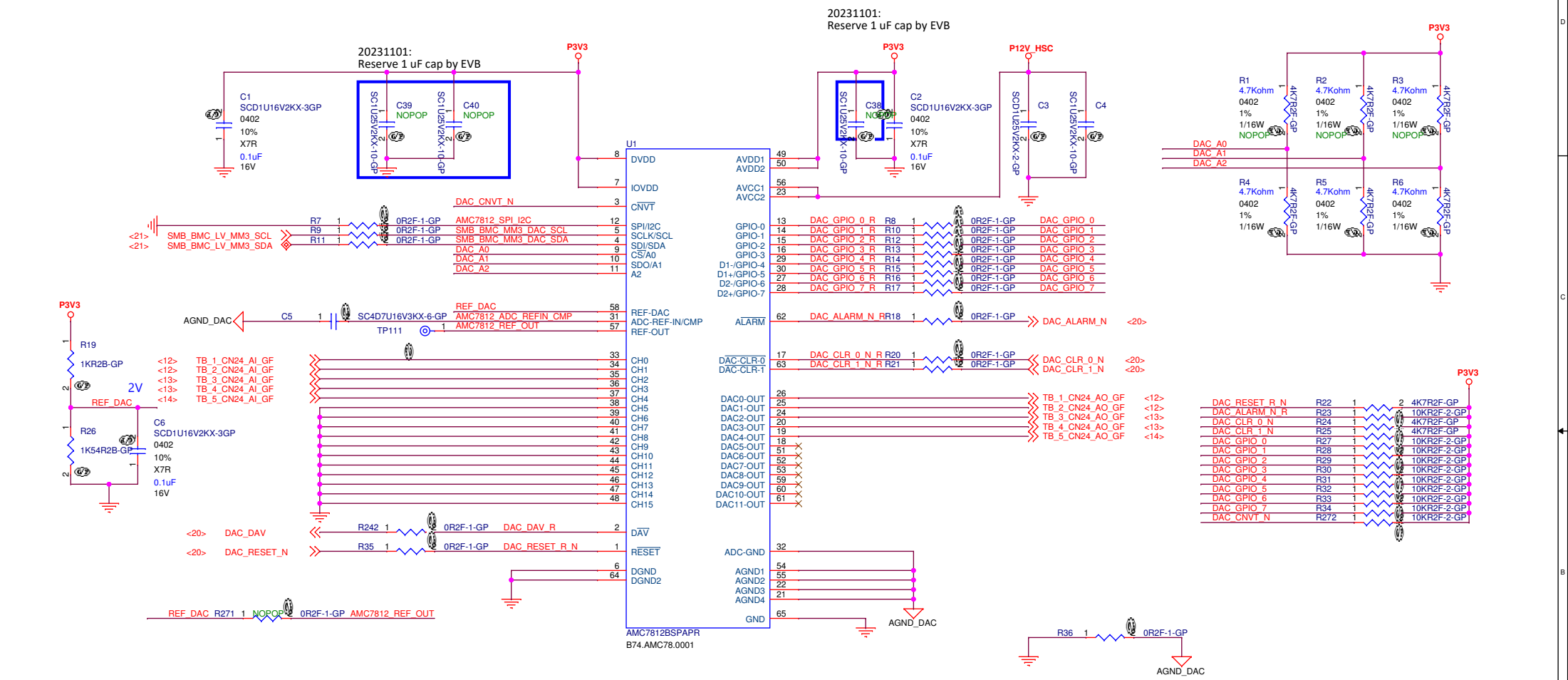
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Conn 3/4 to terminal block board

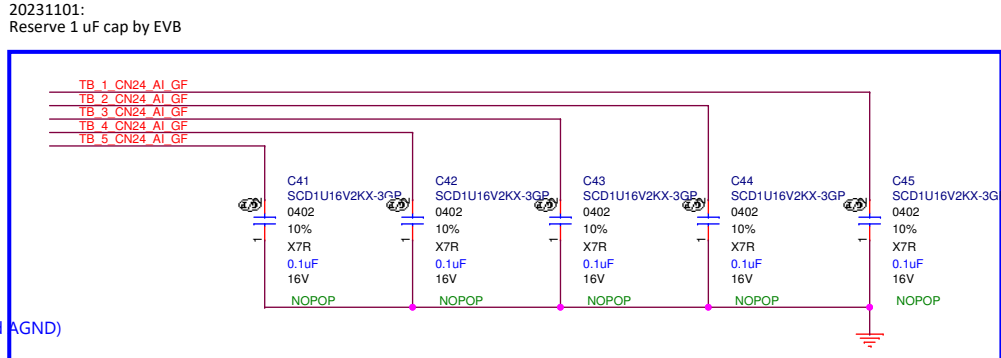


DAC AMC7812BSPAPR

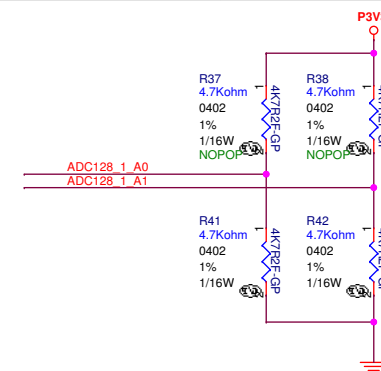
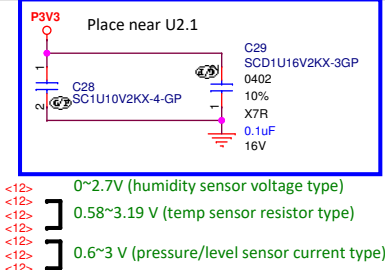
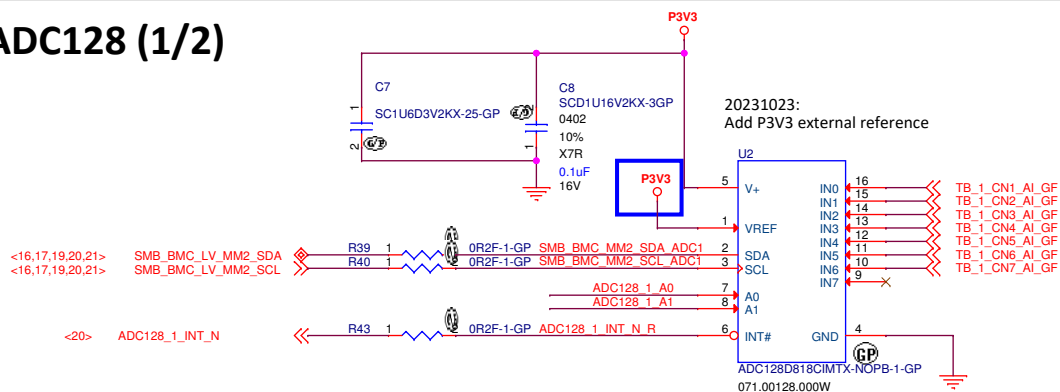


SPI/I2C
Interface selection pin; digital input.
When this pin is tied to IOV DD, the SPI is enabled and the I2C interface is disabled.
When this pin is tied to ground, the SPI is disabled and the I2C interface is enabled.

ADC-REF-IN/CMP 31
External ADC reference input when external V REF is used to drive the ADC.
A compensation capacitor connection(connect a 4.7-μF capacitor between this pin and AGND)
when internal V REF is used to drive the ADC.

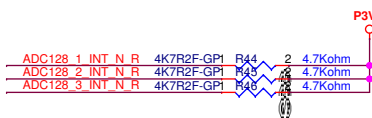
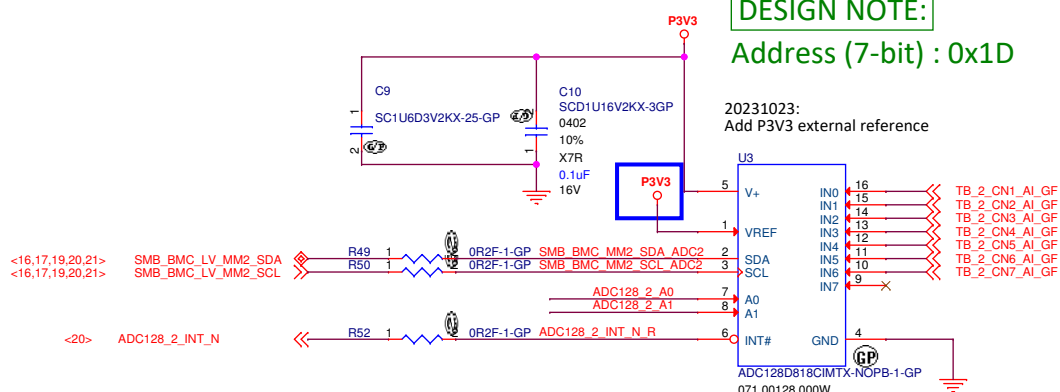


ADC128 (1/2)



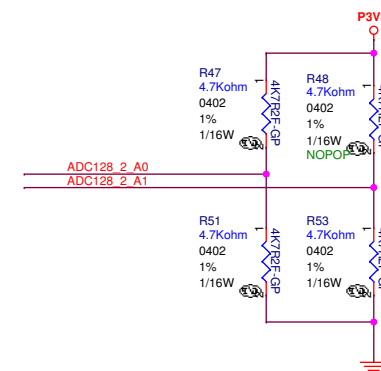
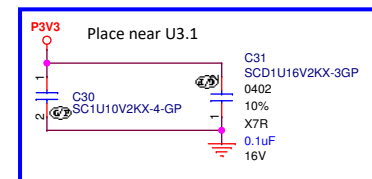
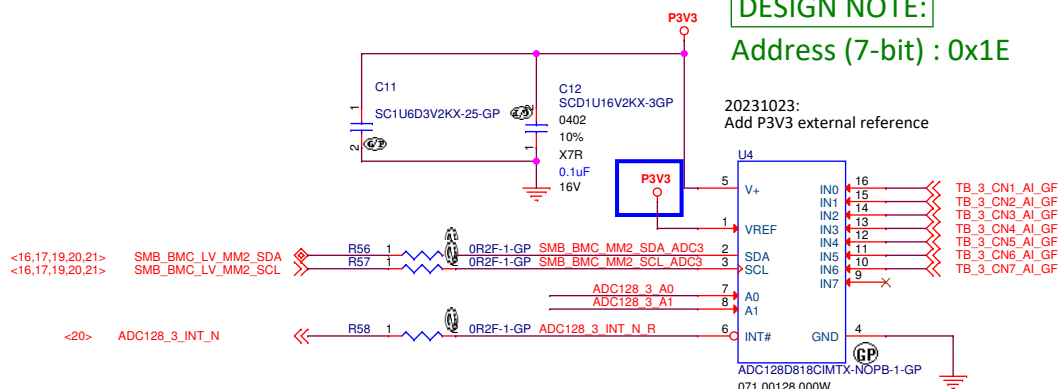
DESIGN NOTE:

Address (7-bit) : 0x1D



DESIGN NOTE:

Address (7-bit) : 0x1E



DESIGN NOTE:

Address (7-bit) : 0x1F

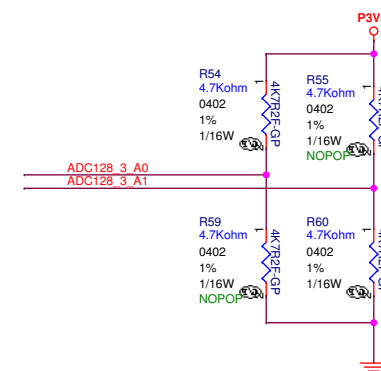
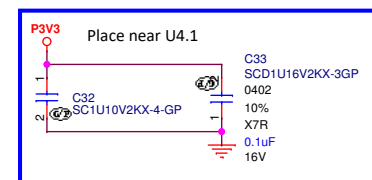
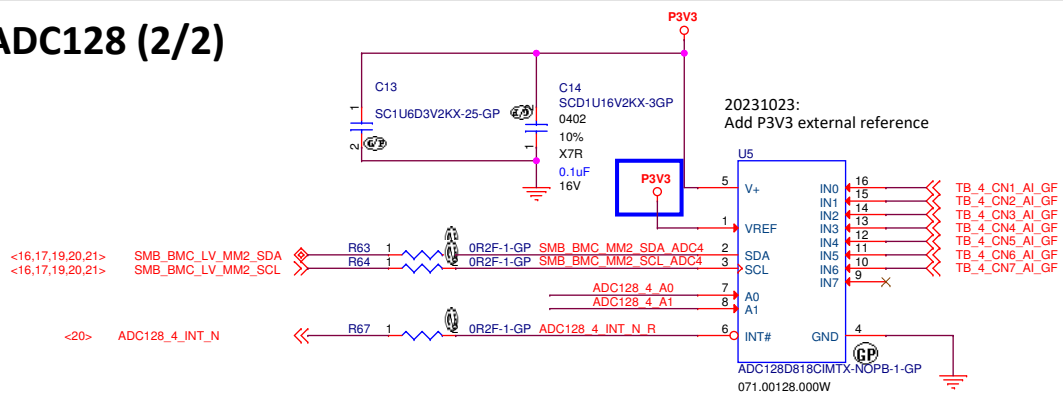


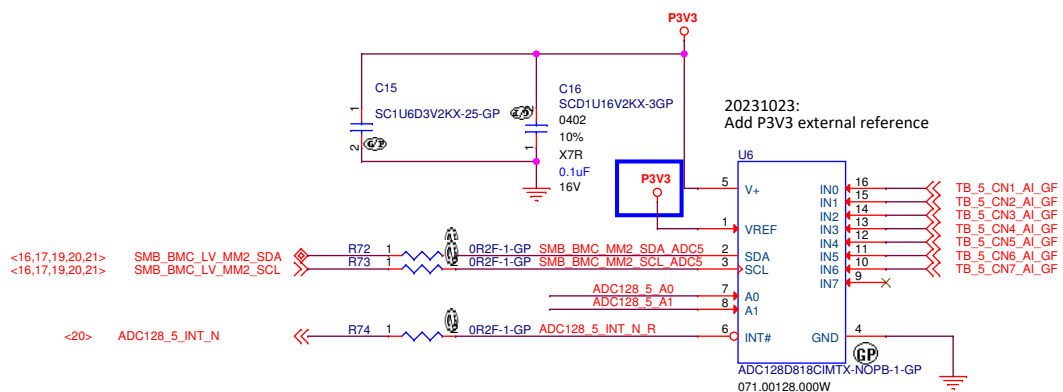
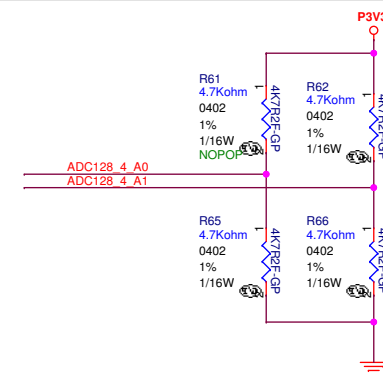
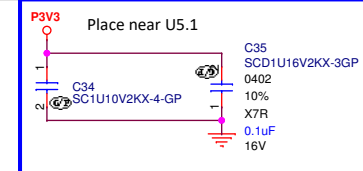
Table 5. Serial Bus Address Table

A1	A0	Serial Bus Address [A0][A1][A2...A9]	Serial Bus Address (hex)
LOW	LOW	001_1101b	1Dh
LOW	MID	001_1110b	1Eh
LOW	HIGH	001_1111b	1Fh
MID	LOW	010_1101b	2Dh
MID	MID	010_1110b	2Eh
MID	HIGH	010_1111b	2Fh
HIGH	LOW	011_0110b	3Dh
HIGH	MID	011_0111b	3Eh
HIGH	HIGH	011_0111b	3Fh

ADC128 (2/2)



DESIGN NOTE:
Address (7-bit) : 0x2D



DESIGN NOTE:
Address (7-bit) : 0x2E

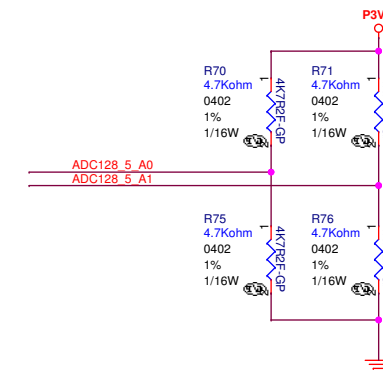
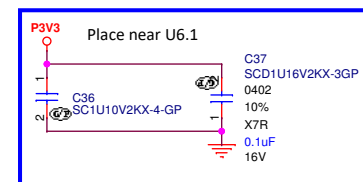
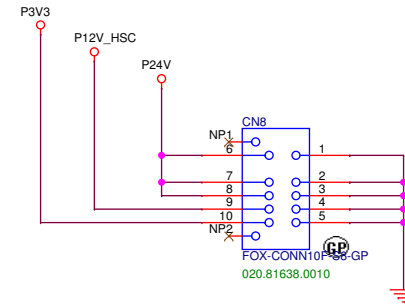
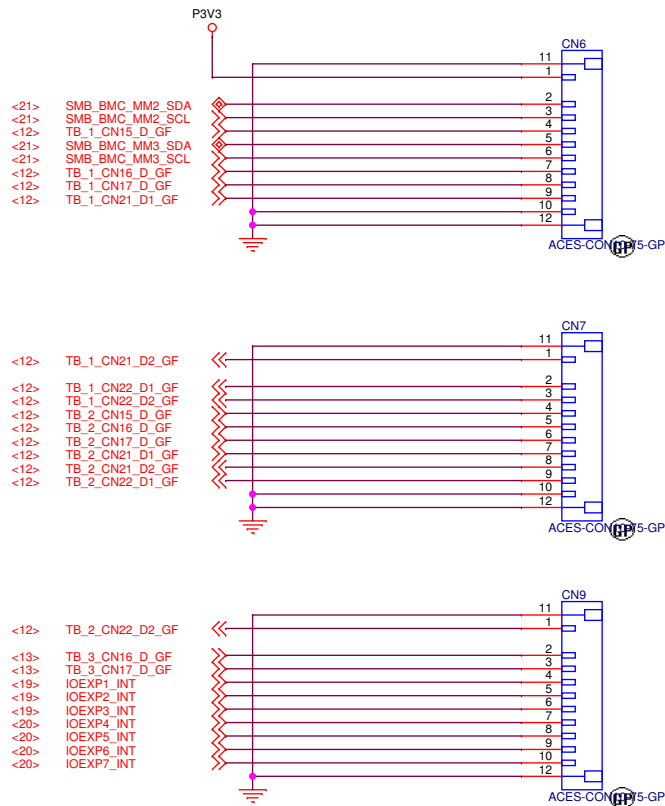


Table 5. Serial Bus Address Table

		Serial Bus Address [A6][A5][A4]..[A0]	Serial Bus Address (hex)
A1	A0	001_1101b	1Dh
LOW	LOW	001_1110b	1Eh
LOW	MID	001_1111b	1Fh
LOW	HIGH	010_1101b	2Dh
MID	LOW	010_1110b	2Eh
MID	MID	010_1111b	2Fh
MID	HIGH	011_0101b	35h
HIGH	LOW	011_0110b	36h
HIGH	MID	011_0111b	37h
HIGH	HIGH		

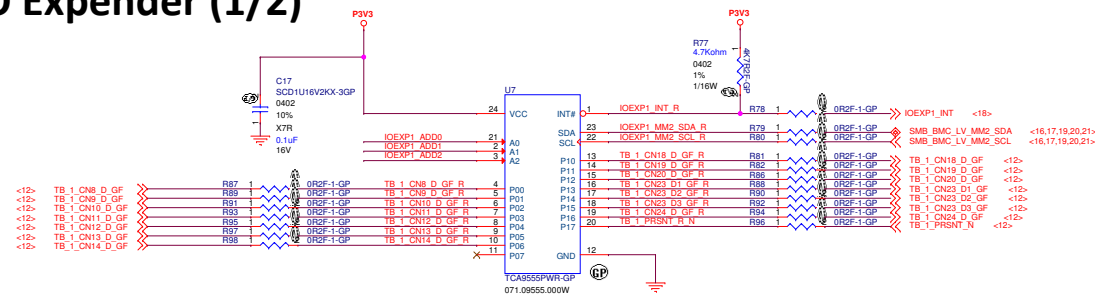
POWER&SIGNAL_CONN



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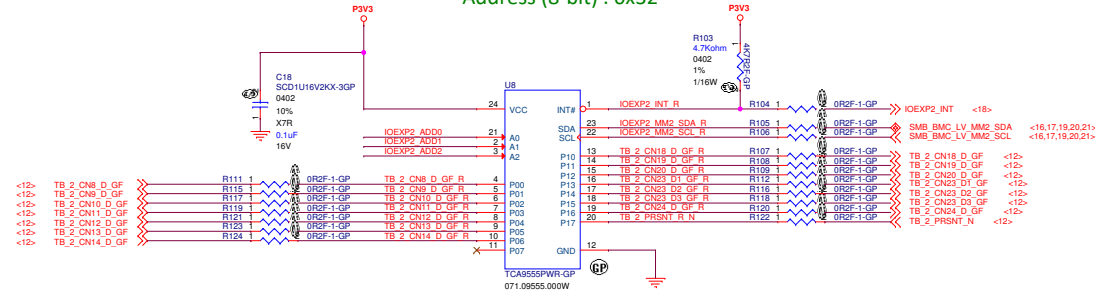
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I/O Expander (1/2)



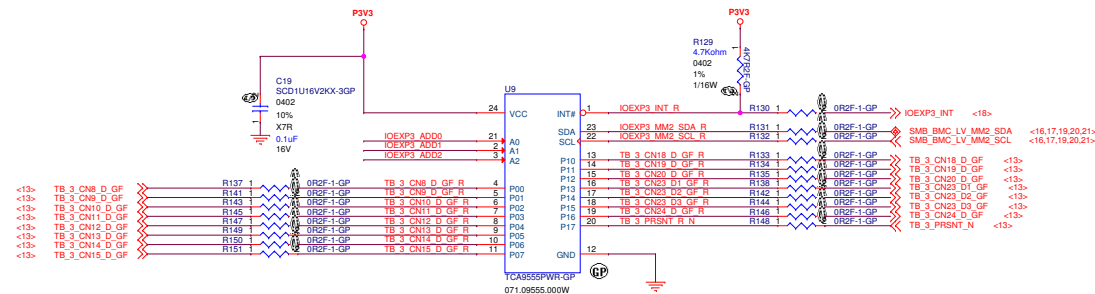
DESIGN NOTE:

Address (7-bit) : 0x20
Address (8-bit) : 0x32



DESIGN NOTE:

Address (7-bit) : 0x21
Address (8-bit) : 0x33



DESIGN NOTE:

Address (7-bit) : 0x22
Address (8-bit) : 0x34

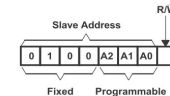
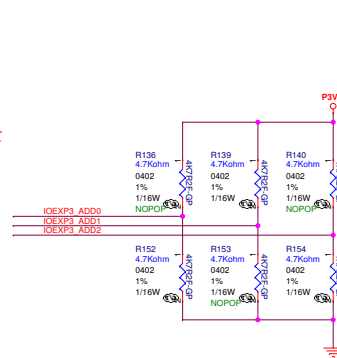
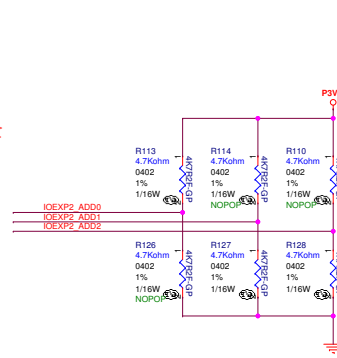
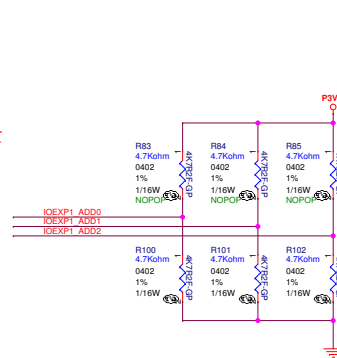
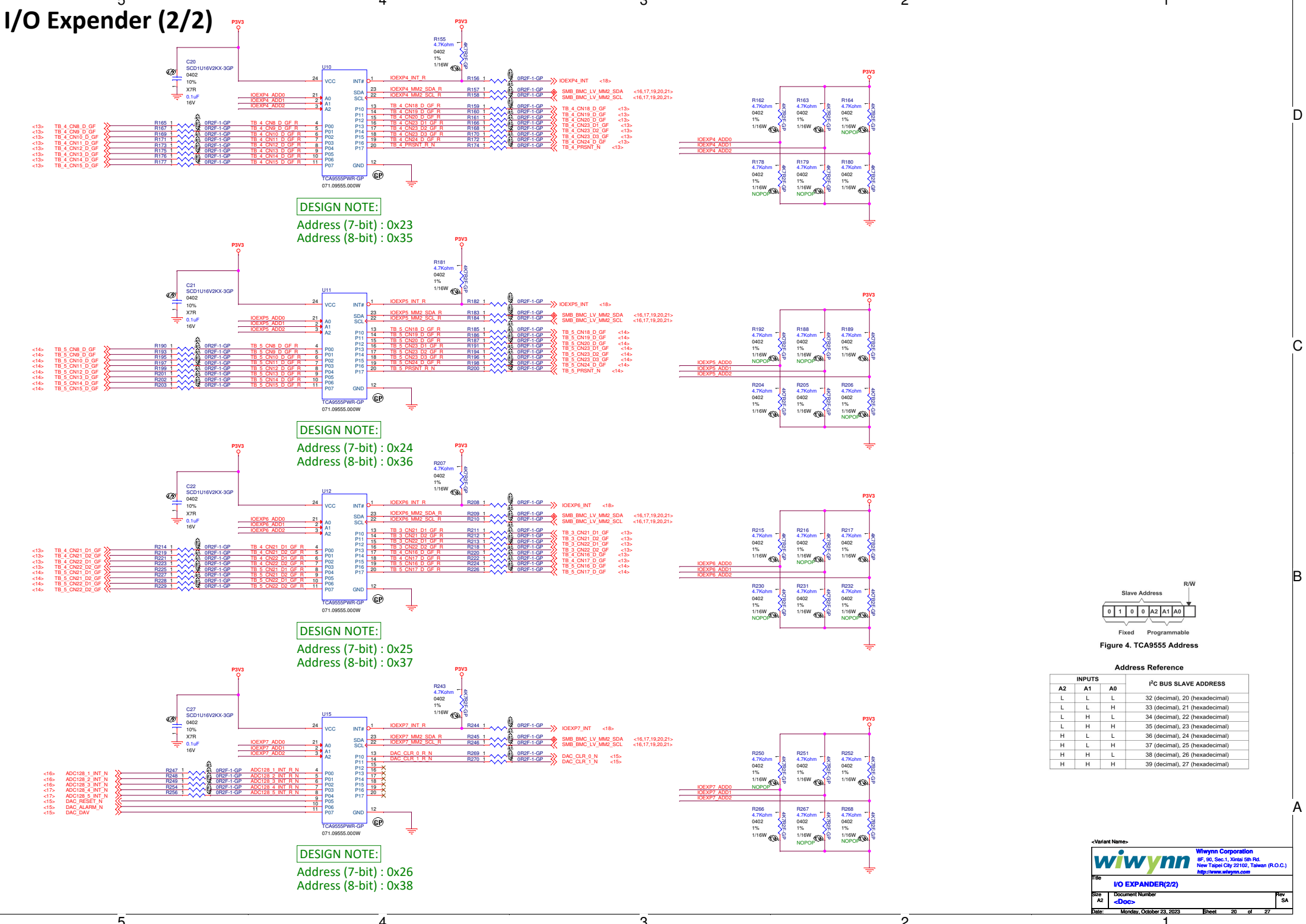


Figure 4. TCA9555 Address

Address Reference

INPUTS			I ² C BUS SLAVE ADDRESS	
A2	A1	A0	L	H
L	L	H	32 (decimal), 20 (hexadecimal)	
L	L	H	33 (decimal), 21 (hexadecimal)	
L	H	L	34 (decimal), 22 (hexadecimal)	
L	H	H	35 (decimal), 23 (hexadecimal)	
H	L	L	36 (decimal), 24 (hexadecimal)	
H	L	H	37 (decimal), 25 (hexadecimal)	
H	H	L	38 (decimal), 26 (hexadecimal)	
H	H	H	39 (decimal), 27 (hexadecimal)	

I/O Expander (2/2)



DESIGN NOTE:
Address (7-bit) : 0x23
Address (8-bit) : 0x35

DESIGN NOTE:
Address (7-bit) : 0x24
Address (8-bit) : 0x36

DESIGN NOTE:
Address (7-bit) : 0x25
Address (8-bit) : 0x37

DESIGN NOTE:
Address (7-bit) : 0x26
Address (8-bit) : 0x38

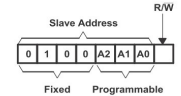
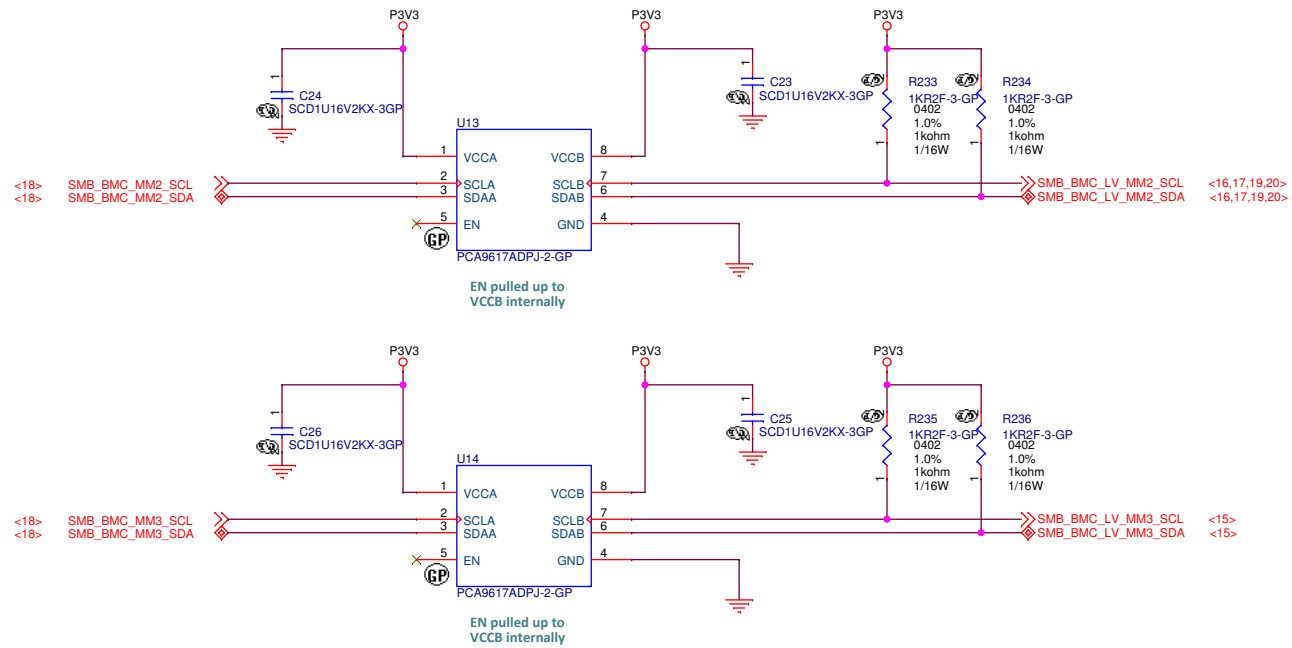


Figure 4. TCA9555 Address

Address Reference			
INPUTS			
A2	A1	A0	I ² C BUS SLAVE ADDRESS
L	L	L	32 (decimal), 20 (hexadecimal)
L	L	H	33 (decimal), 21 (hexadecimal)
L	H	L	34 (decimal), 22 (hexadecimal)
L	H	H	35 (decimal), 23 (hexadecimal)
H	L	L	36 (decimal), 24 (hexadecimal)
H	L	H	37 (decimal), 25 (hexadecimal)
H	H	L	38 (decimal), 26 (hexadecimal)
H	H	H	39 (decimal), 27 (hexadecimal)

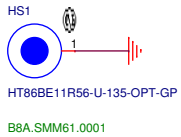
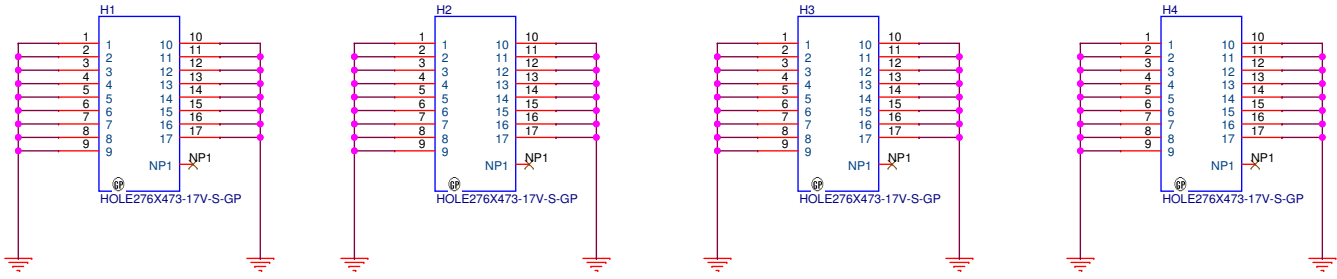
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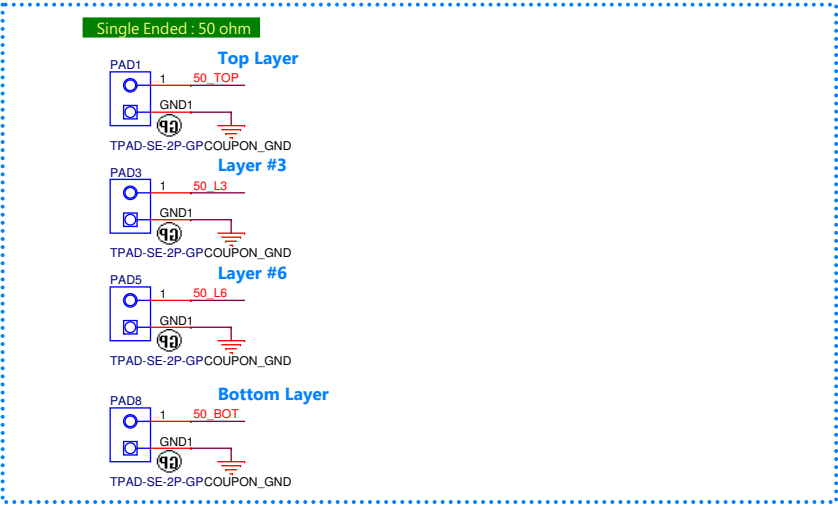
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SCREW HOLE



IMPEDANCE COUPON



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DELTA-L 4.0 COUPON

CHANGE HISTORY

Version	Item	Date	Page	DESCRIPTION	BOM change	Schematic change	Layout change	Reason
1	1	2023/10/23	16, 17	Add 3.3V external power reference on U2~U6	✓	✓	✓	To extend sensor reading range
1	2	2023/11/01	15	Reserve 1 uF cap by EVB	✓	✓	✓	Reserve 1 uF cap by EVB